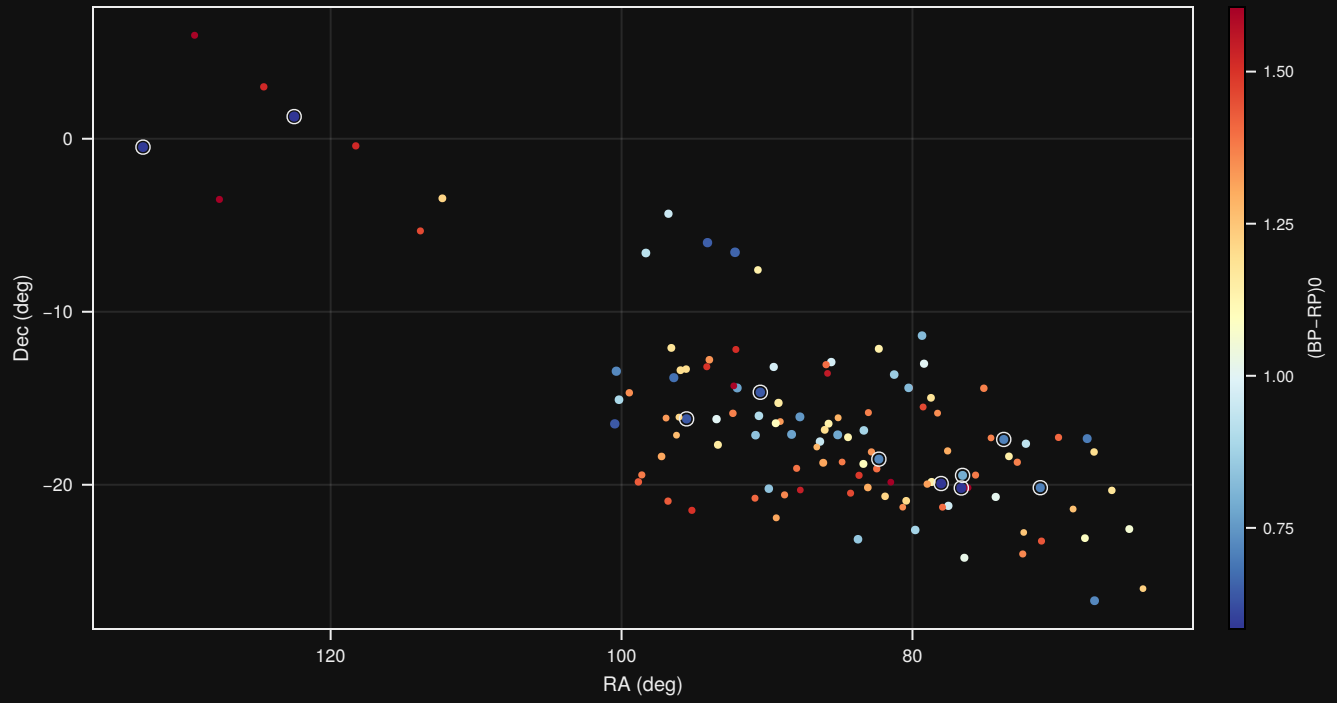
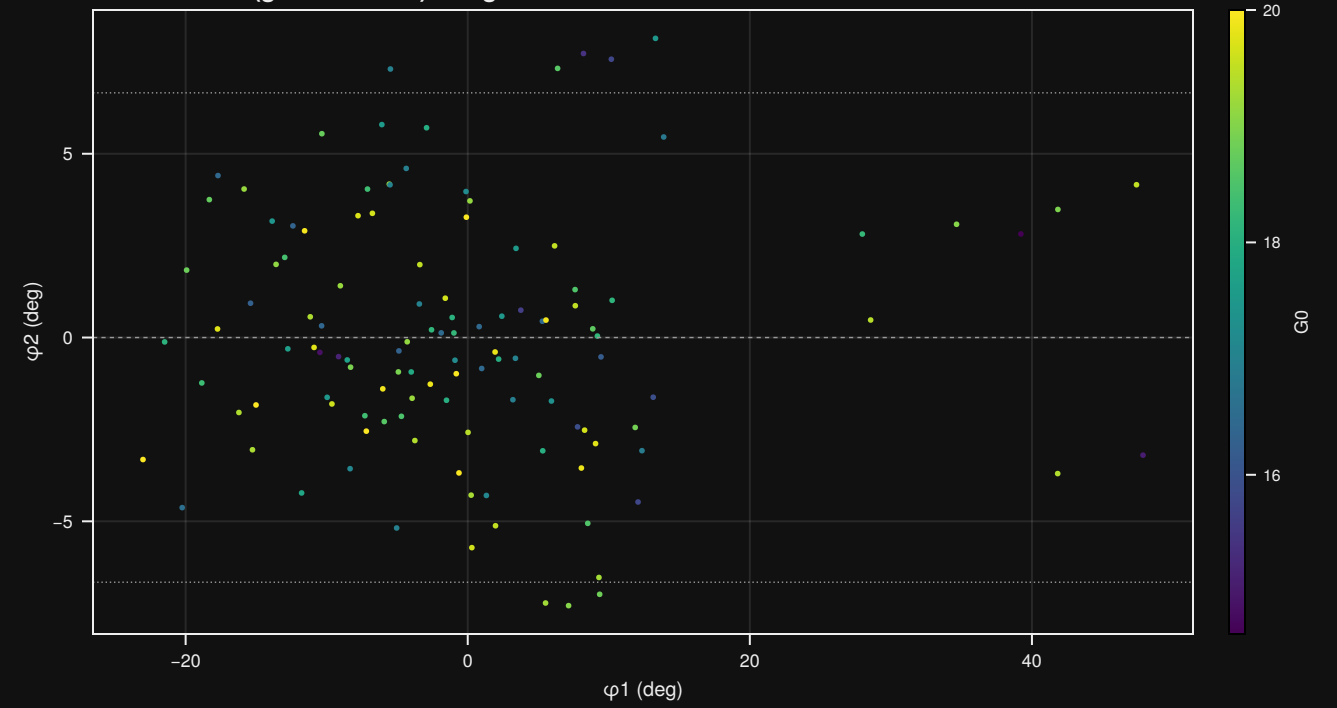
New-3

MEMBERSHIP	
Stars in catalog	120
with Gaia GSP-Phot	63 (52%)
with radial velocity	10
Stream id (sID)	17
GEOMETRY · great-circle fit	
Centroid RA, Dec	86.8, -17.2 deg
Centroid l, b	221.2, -22.6 deg
Pole RA, Dec	36.4, 64.8 deg
Length (φ1, 2–98%)	62.0 deg
Width σ(φ2), robust	3.33 deg
KINEMATICS	
pmRA* med ± σ	-29.10 ± 7.69 mas/yr
pmDE med ± σ	-25.53 ± 5.83 mas/yr
μφ1* med ± σ	-35.84 ± 9.60 mas/yr
μφ2 med ± σ	-14.59 ± 2.39 mas/yr
Parallax (cat.) med	0.635 ± 0.184 mas
Parallax (wtd, ZP-corr.)	0.644 ± 0.013 mas
Distance	d = 1.6 ± 0.0 kpc
Physical length	1.68 kpc
RV median ± σ(MAD)	263.6 ± 34.2 km/s
RV gradient dV/dφ1	-4.90 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	14.6–20.0 (18.5)
(BP−RP)0 median	1.20
GSP-Phot [M/H] (n=63)	-2.42 ± 0.63
GSP-Phot Teff median	5144 K
RUWE > 1.4	0.0 %
PM clusters (largest)	10 (16 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	120 | 120
n_RV catalog | paper	10 | 10
<[Fe/H]> paper	-2.71
Stellar mass (paper)	120 Msun
Pole offset vs paper	15.6 deg
Remark	Two clumps?

Sky map · catalog members (n = 120) · size ∝ brightness · ring = has RV

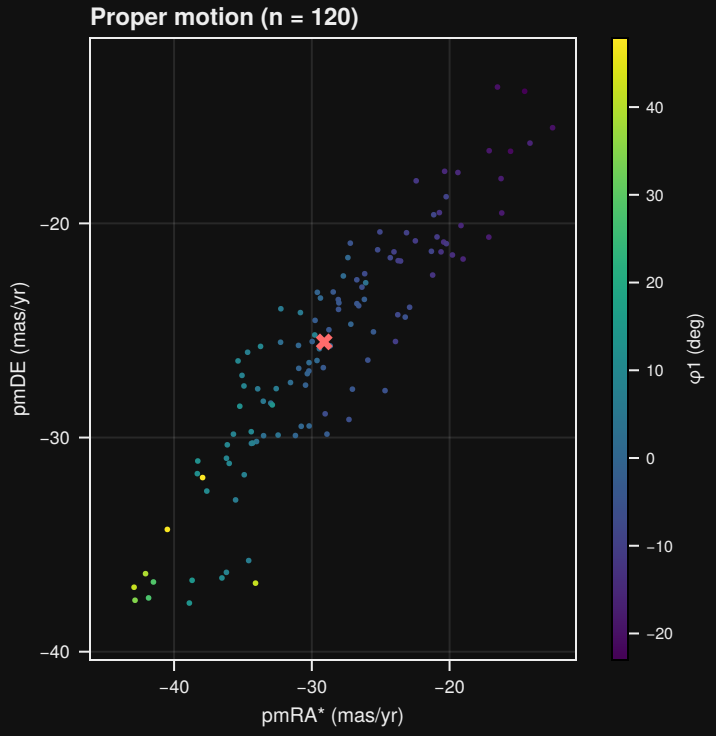


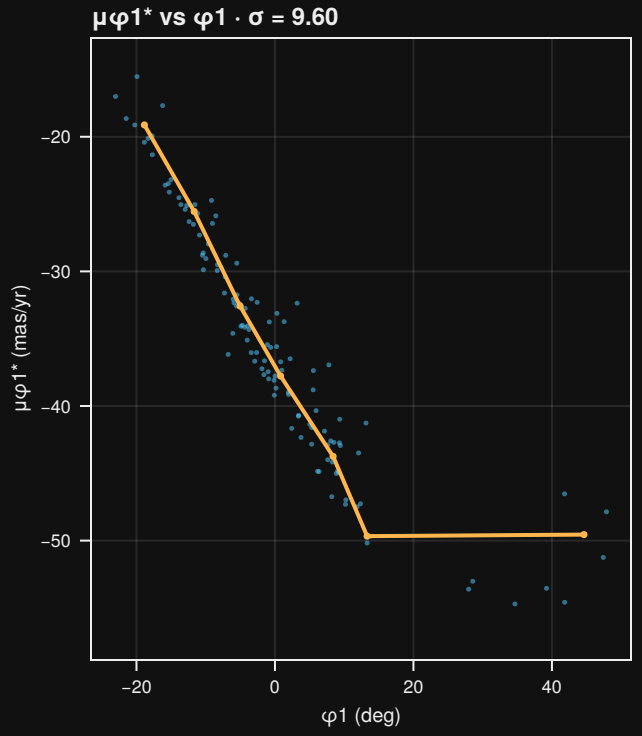
Stream frame (great-circle fit) · length 62.0° · width σ 3.33°

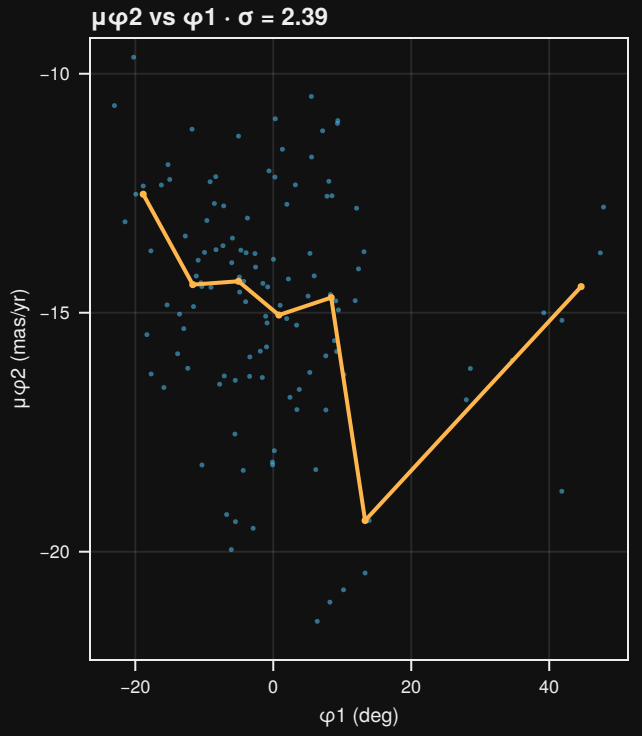


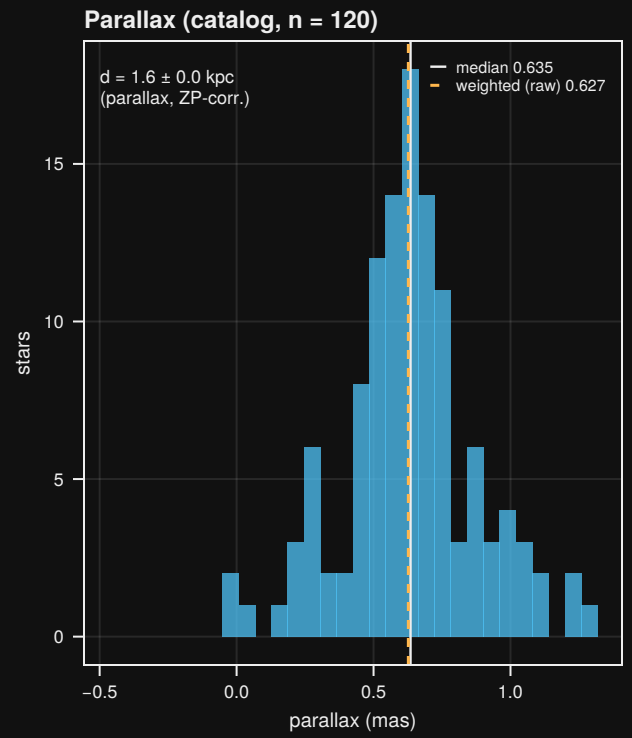
New-3

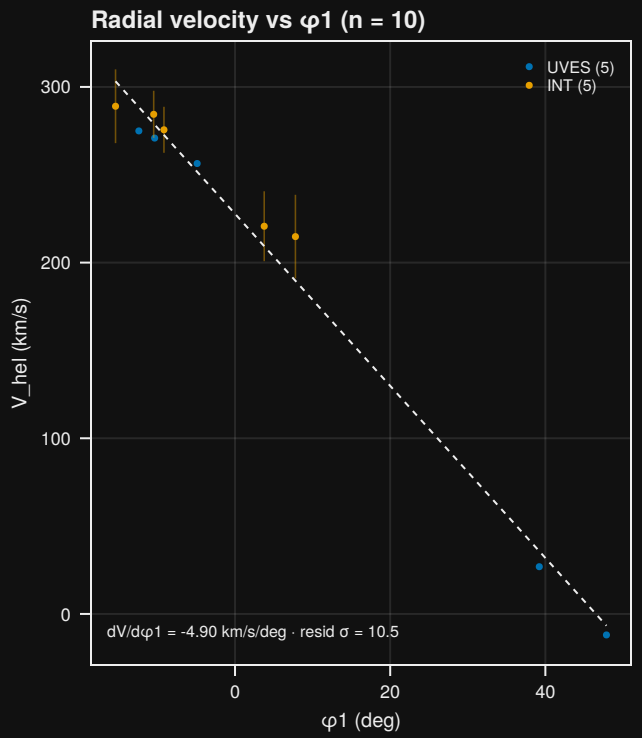
Kinematics · proper motion · parallax · radial velocity

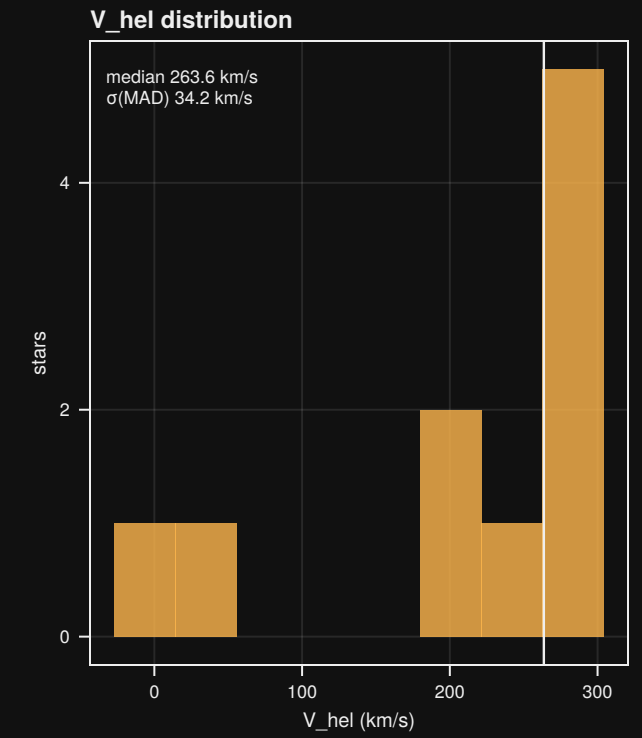






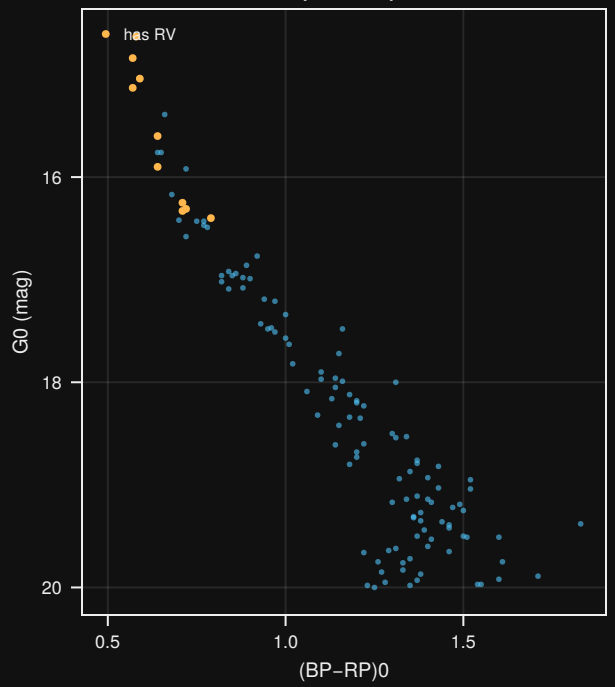




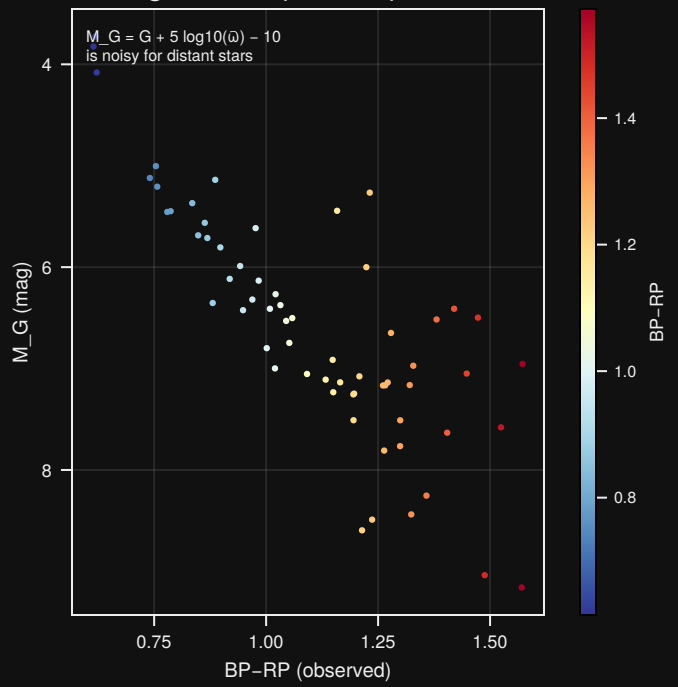


New-3

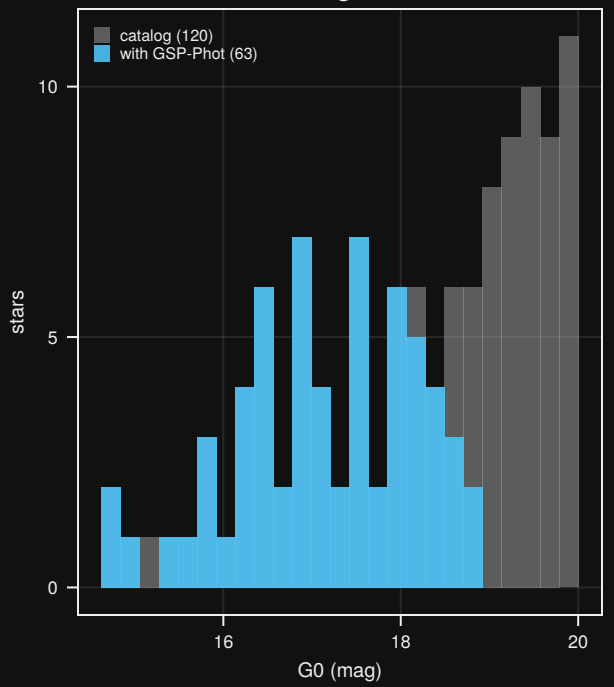
CMD · dereddened (n = 120)



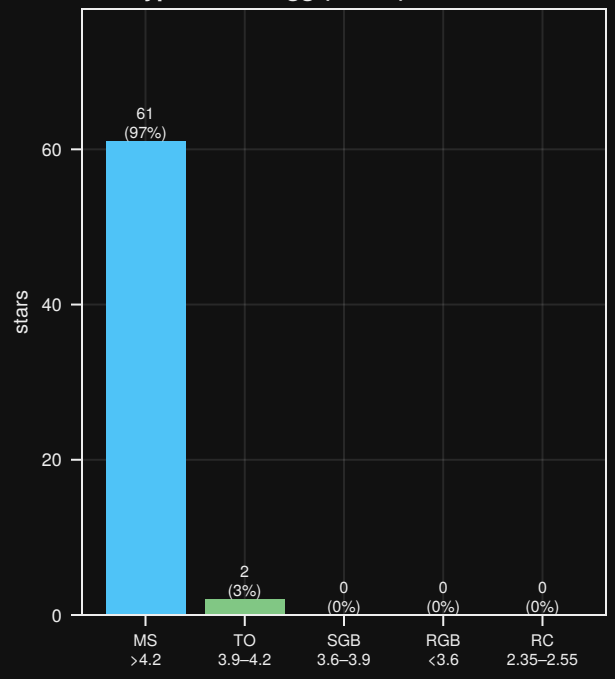
HR diagram · ϖ>0 (n = 63/63)



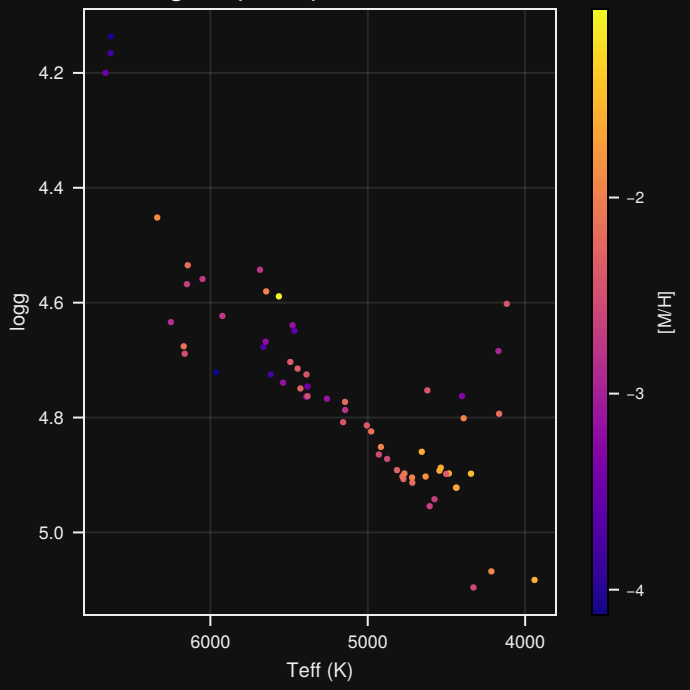
G0 & GSP-Phot coverage



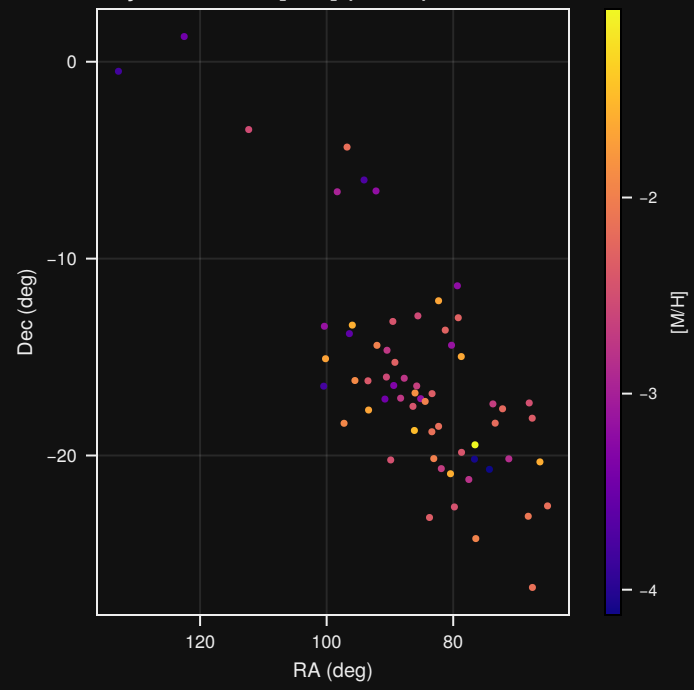
Star types from logg (n = 63)



Kiel diagram (n = 63)



Sky · GSP-Phot [M/H] (n = 63)

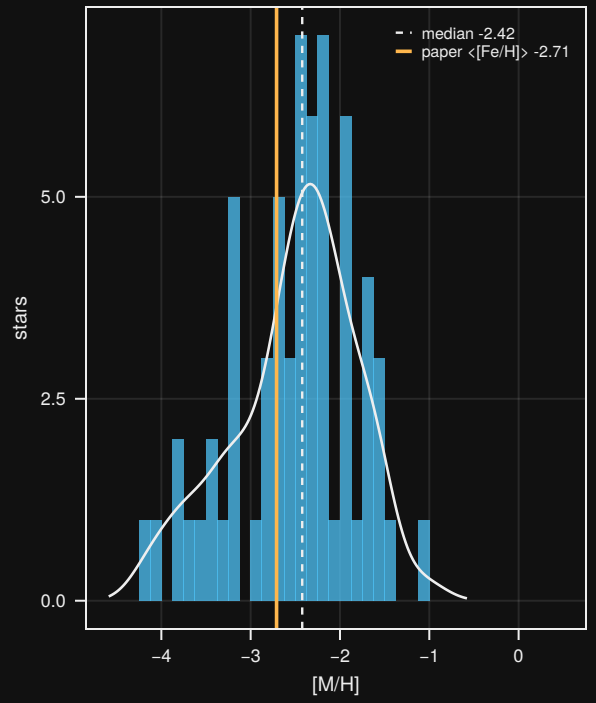


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

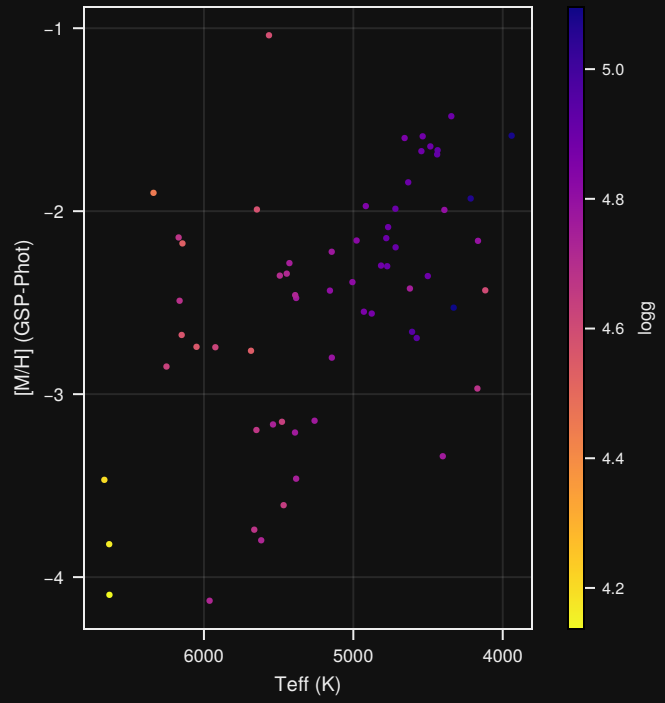
New-3

Metallicity · age · astrometric quality

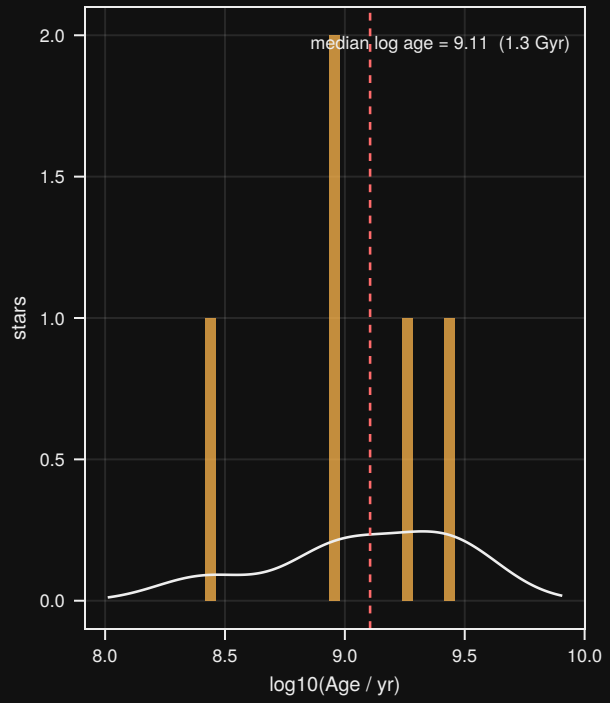
[M/H] GSP-Phot (n = 63)



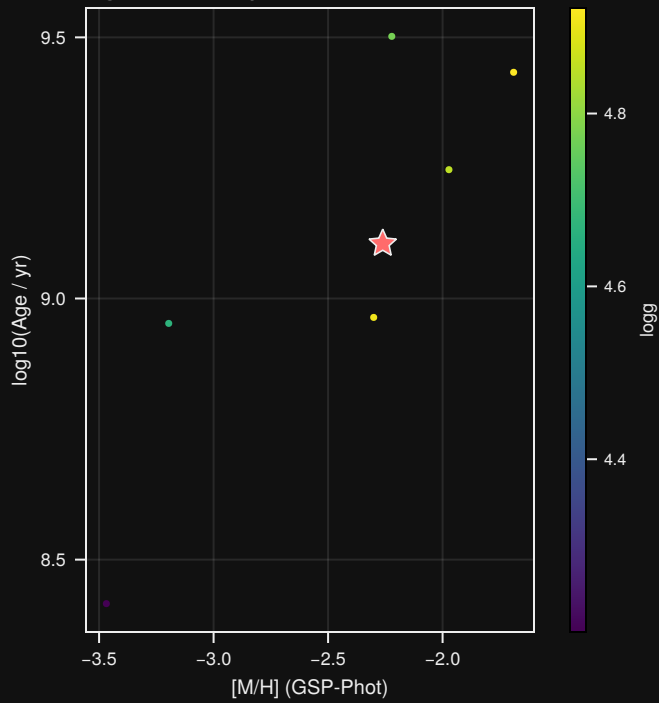
Teff vs [M/H] (n = 63)



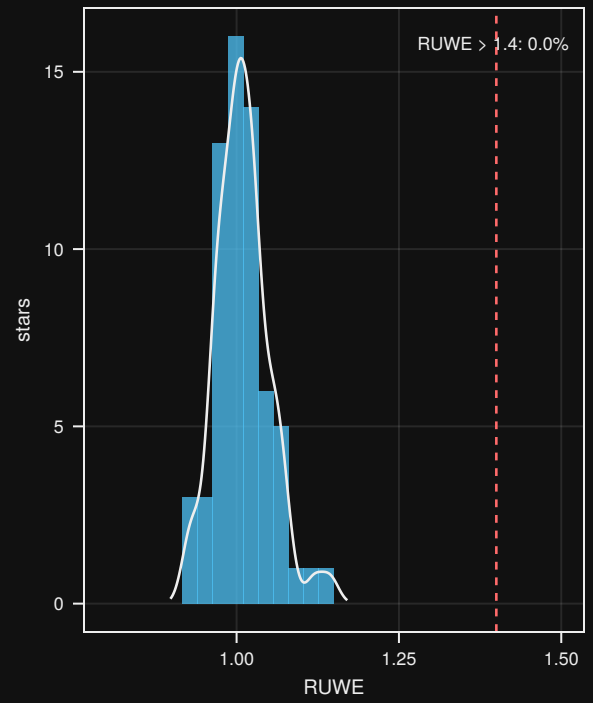
FLAME age (n = 6)



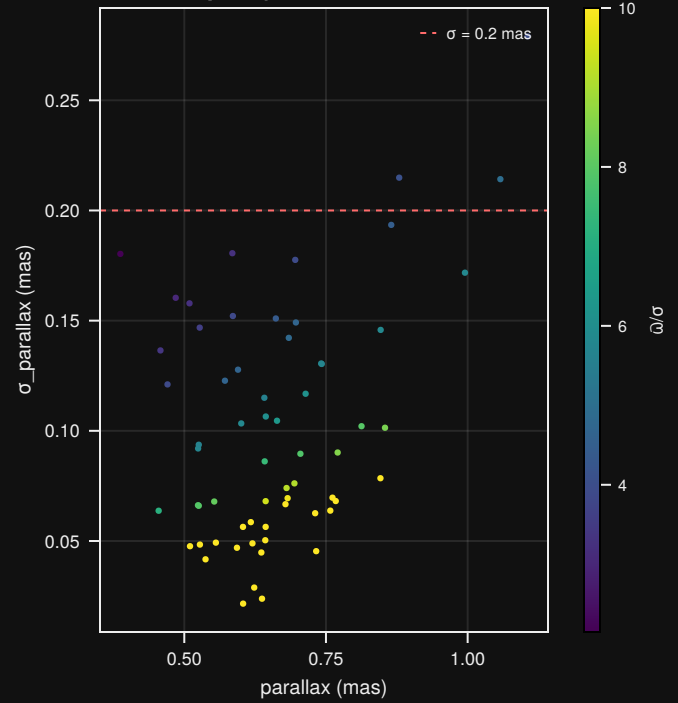
Age–metallicity (n = 6)



RUWE (n = 63)

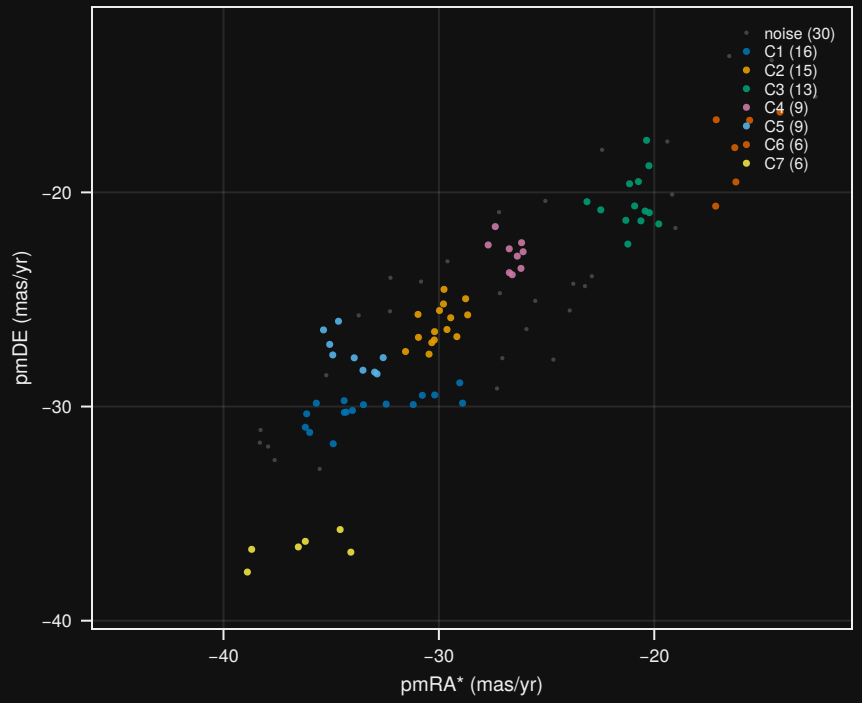


Parallax quality (n = 63)

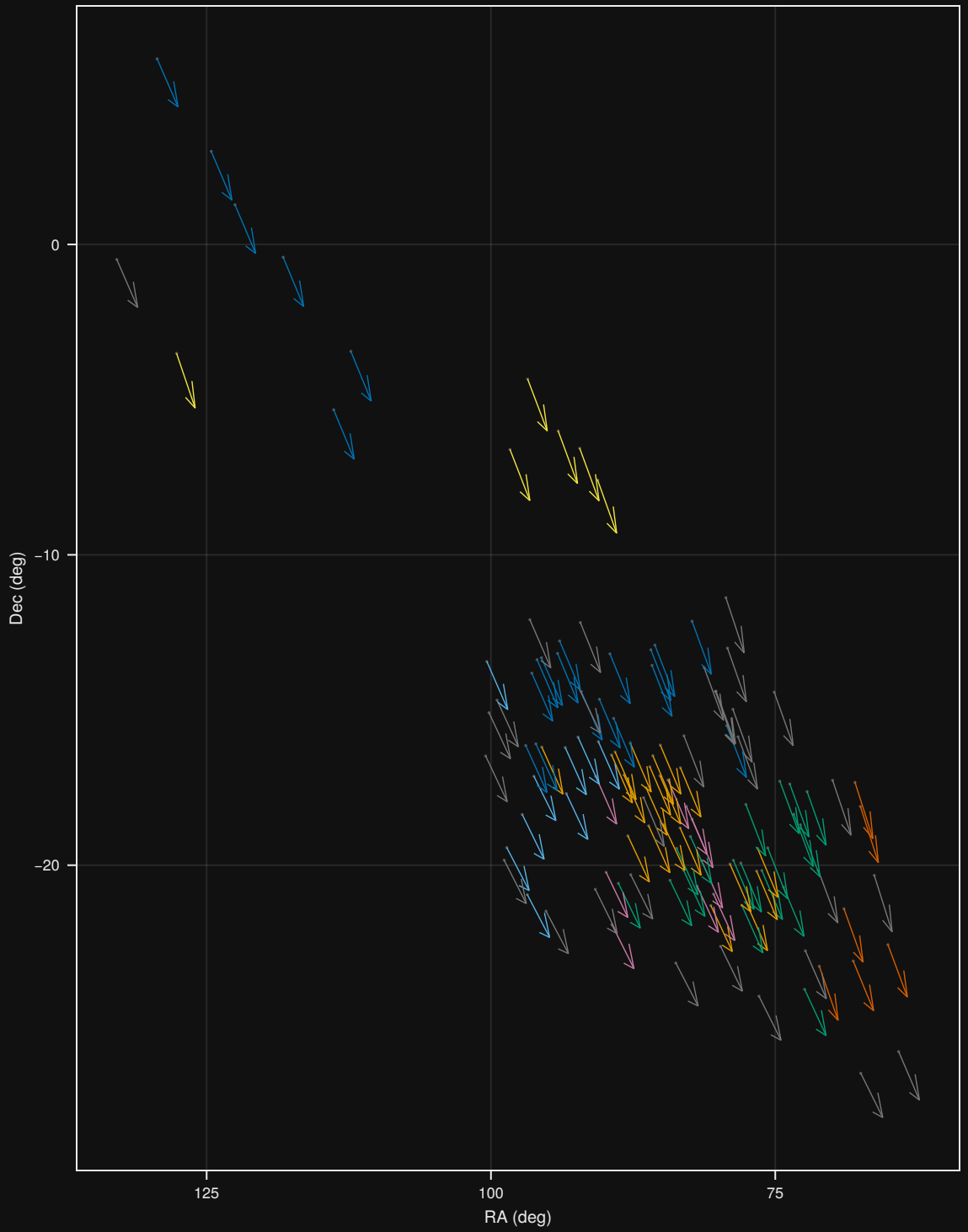


New-3

PM clustering · HDBSCAN · 10 cluster(s) · noise 25%



Proper-motion directions (arrows) coloured by cluster



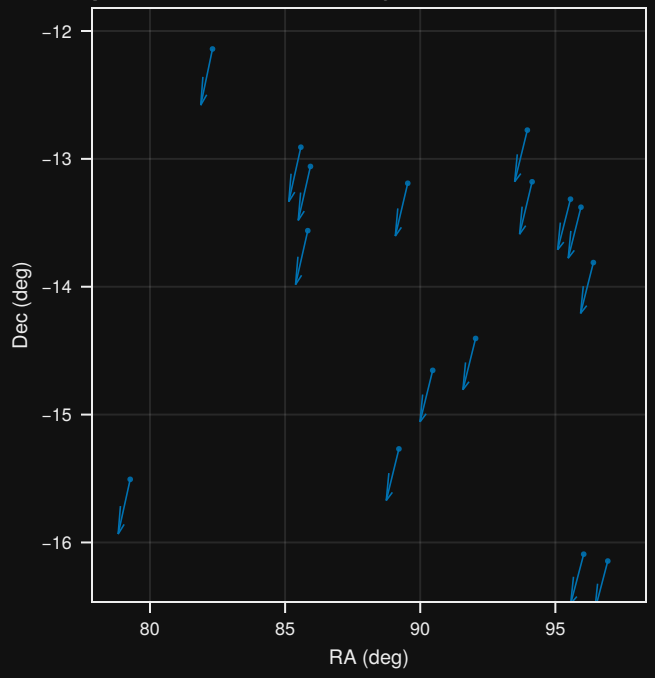
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	30	-26.49	-24.54	7.48	-2.32	2
C1	16	-34.16	-29.91	2.63	-2.37	1
C2	15	-29.97	-26.40	0.72	-2.43	1
C3	13	-20.74	-20.82	0.72	-2.42	2
C4	9	-26.59	-22.77	0.60	-2.33	1
C5	9	-33.93	-27.72	1.48	-2.39	0
C6	6	-16.24	-17.27	1.15	-2.26	0
C7	6	-36.36	-36.61	3.01	-3.08	0
C8	6	-41.96	-36.87	0.99	-3.00	1

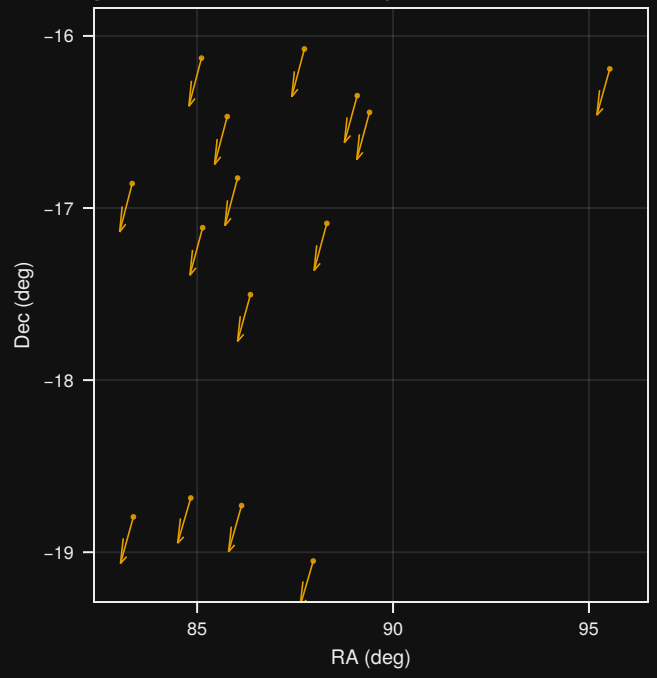
New-3

Per-cluster proper-motion vector maps

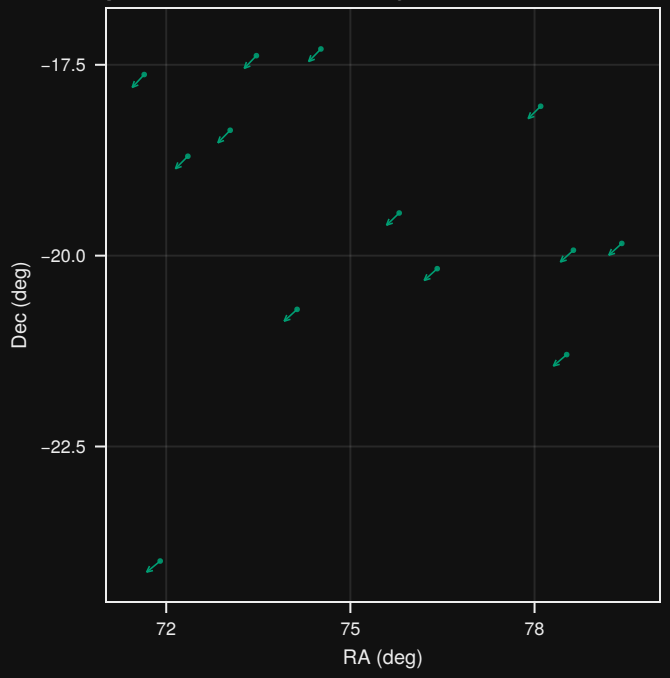
Cluster C1 · n = 16
pm = (-34.16, -29.91) mas/yr



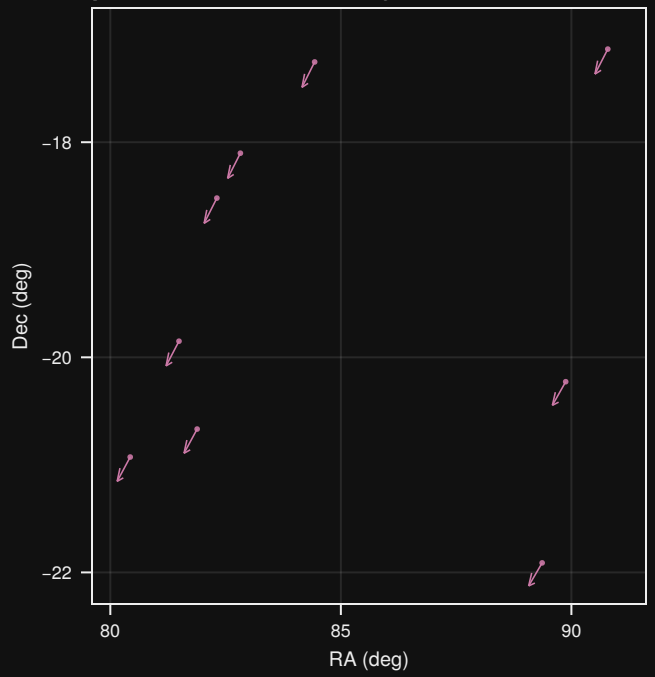
Cluster C2 · n = 15
pm = (-29.97, -26.40) mas/yr



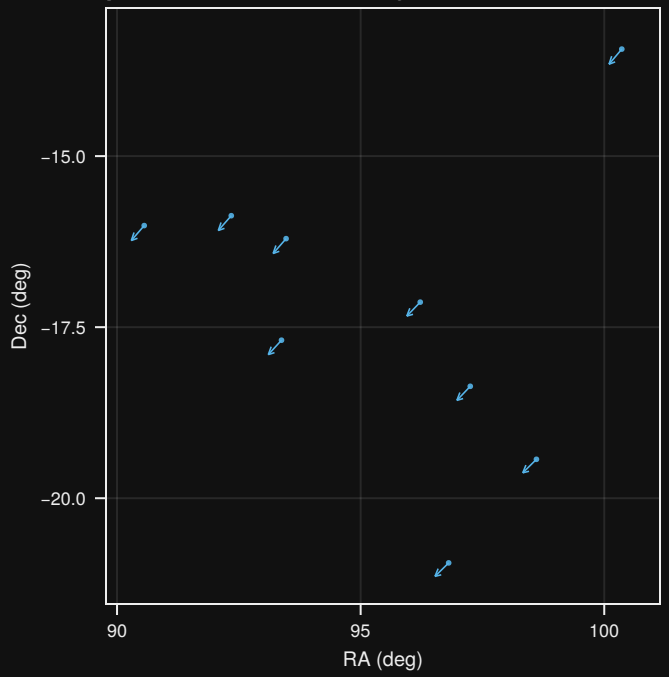
Cluster C3 · n = 13
pm = (-20.74, -20.82) mas/yr



Cluster C4 · n = 9
pm = (-26.59, -22.77) mas/yr



Cluster C5 · n = 9
pm = (-33.93, -27.72) mas/yr



Noise · n = 30

